

Matplotlib for Data Science

Detailed Notes with Complete Code

1. Introduction to Matplotlib

Matplotlib is a Python library used for creating **2D data visualizations**.

In data science, it is mainly used during **Exploratory Data Analysis (EDA)** to:

- Understand data distribution
- Identify patterns and trends
- Detect outliers
- Compare multiple variables
- Present insights visually

Matplotlib works seamlessly with **NumPy arrays** and **Pandas DataFrames**, which makes it a core visualization tool in the data science workflow.

2. Importing Matplotlib and Basic Setup

The most commonly used module in Matplotlib is `pyplot`.

```
import matplotlib.pyplot as plt
```

The typical workflow of any Matplotlib plot:

1. Prepare data
2. Create the plot

3. Add labels and title
4. Display the plot

3. Line Plot

Purpose

A line plot is used to show **trends over a continuous variable**, usually time.

Data Science Use Cases

- Stock price movement
- Growth or decline over time
- Sensor or log data analysis

Code Example

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [10, 20, 25, 30, 40]

plt.plot(x, y)
plt.xlabel("Time")
plt.ylabel("Value")
plt.title("Line Plot Example")
plt.show()
```

Explanation

- `plt.plot(x, y)` draws a line connecting the data points
- `xlabel`, `ylabel`, and `title` improve readability

- `plt.show()` displays the plot

4. Bar Chart

Purpose

Bar charts are used to **compare values across categories**.

Data Science Use Cases

- Sales by product
- Number of students in each department
- Frequency comparison of categories

Code Example

```
import matplotlib.pyplot as plt

categories = ['A', 'B', 'C', 'D']
values = [15, 30, 45, 10]

plt.bar(categories, values)
plt.xlabel("Category")
plt.ylabel("Count")
plt.title("Bar Chart Example")
plt.show()
```

Explanation

Each bar represents a category, and its height represents the corresponding value.

5. Histogram

Purpose

A histogram is used to understand the **distribution of numerical data**.

Data Science Importance

This is one of the **most important plots in EDA**.

It helps to identify:

- Distribution shape
- Data spread
- Skewness
- Outliers

Code Example

```
import matplotlib.pyplot as plt
import numpy as np

data = np.random.normal(50, 10, 100)

plt.hist(data, bins=10)
plt.xlabel("Value Range")
plt.ylabel("Frequency")
plt.title("Histogram Example")
plt.show()
```

Explanation

- `bins` controls how many intervals the data is divided into
- A higher number of bins shows more detail

6. Scatter Plot

Purpose

Scatter plots are used to **analyze the relationship between two numerical variables**.

Data Science Use Cases

- Correlation analysis
- Feature relationship analysis
- Outlier detection

Code Example

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [5, 7, 6, 8, 10]

plt.scatter(x, y)
plt.xlabel("Independent Variable")
plt.ylabel("Dependent Variable")
plt.title("Scatter Plot Example")
plt.show()
```

Explanation

Each point represents a pair of values from the dataset.

7. Box Plot

Purpose

Box plots show the **statistical summary of data**.

What It Displays

- Minimum value

- First quartile (Q1)
- Median
- Third quartile (Q3)
- Maximum value
- Outliers

Code Example

```
import matplotlib.pyplot as plt

data = [10, 12, 15, 18, 19, 21, 22, 100]

plt.boxplot(data)
plt.ylabel("Values")
plt.title("Box Plot Example")
plt.show()
```

Explanation

Outliers are shown as individual points beyond the whiskers.

8. Subplots

Purpose

Subplots allow multiple graphs to be displayed in **a single figure**.

Data Science Use Cases

- Comparing different visualizations
- Side-by-side analysis

Code Example

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4]  
y1 = [10, 20, 30, 40]  
y2 = [40, 30, 20, 10]
```

```
plt.subplot(1, 2, 1)  
plt.plot(x, y1)  
plt.title("Plot 1")
```

```
plt.subplot(1, 2, 2)  
plt.plot(x, y2)  
plt.title("Plot 2")
```

```
plt.show()
```

Explanation

- `subplot(rows, columns, index)`
- Both plots appear in one figure

9. Customization and Styling

Adding Grid

```
plt.grid(True)
```

Changing Line Style and Color

```
plt.plot(x, y, color='red', linestyle='--')
```

Importance

Customization improves **clarity and presentation**, especially in reports and projects.

10. Working with Pandas DataFrames

Matplotlib is tightly integrated with Pandas.

Example

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Marks': [55, 65, 70, 85, 90]
}

df = pd.DataFrame(data)

df['Marks'].plot(kind='hist')
plt.xlabel("Marks")
plt.title("Marks Distribution")
plt.show()
```

Explanation

This method is commonly used in real-world datasets.

11. Saving Plots

Purpose

Saving plots is essential for:

- Assignments
- Reports
- Presentations

Code Example

```
plt.plot(x, y)
plt.savefig("output_plot.png")
plt.show()
```